7			
	(a)	(magnetic) force (always) normal to velocity/direction of motion B1 so provides the <u>centripetal force</u> or a <u>force towards the centre of a circle</u> B1 <u>magnitude</u> of (magnetic) force constant (accept omission of magnitude if direction of force mentioned above) or speed is constant/kinetic energy is constant B1	[3]

	(b)	increase in KE = loss in PE **or** ½ $mv^2 = qV$ KE = $p^2/2m$ or $p = mv$ with algebra leading to $p = \sqrt{(2mqV)}$	M1 A1	[2]
	(c)	Magnetic force provides the centripetal force $Bqv = mv^2 / r$ $mv = Bqr$ **or** $p = Bqr$ $(2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 120)^{1/2} = B \times 1.60 \times 10^{-19} \times 0.074$ $B = 5.0 \times 10^{-4}$ T	 M1 C1 A1	[3]

8	**(a)**	**(i)**	To remain at rest, net force = 0	[1]